

Arranging circles of radii $1, 2, \dots, n$ around a central circle: a Supnick TSP and a geometric breakdown cascade

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Abstract

Non-overlapping circles of radii $1, 2, \dots, n$ must all be externally tangent to a central circle. In what cyclic order should they be arranged so that the radius R of the central circle is minimized? The question was posed by Dan (Mathematics Stack Exchange, January 2023) together with a conjectured optimal arrangement, and is listed as problem 295 of the UQ benchmark of unsolved questions. We resolve the chain-ordering component of the problem and document a structured breakdown cascade. First, we show that the angular-separation matrix $\theta_R(a, b) = 2 \arcsin \sqrt{ab/((R+a)(R+b))}$ is, for every R , a symmetric anti-Monge (inverse Supnick) matrix; by a classical theorem of Supnick (1957) the cyclic order minimizing the total angle is therefore a single fixed tour, independent of R , namely $\sigma^* = \langle 1, n-1, 3, n-3, 5, \dots, n-2, 2, n \rangle$ — an identification that holds for arbitrary *distinct* positive radii, not only $1, \dots, n$ (ties are handled by a limiting argument). This proves the conjectured “pyramid” arrangement optimal whenever the resulting necklace is geometrically realizable, and yields the unconditional lower bound $R^*(n) \geq R_{\text{chain}}(\sigma^*)$. Second, realizability fails in a structured cascade: from $n = 8$ the smallest circle can no longer separate its two large neighbours and becomes a *floating* circle; from $n = 13$ the same mechanism reaches circle 2 inside the chain. We formalize the problem as a circular system of pairwise angular constraints (a simple temporal network), give an exact certified branch-and-bound, and determine the optimal arrangements for all $n \leq 14$, with optima verified to 50 decimal digits. The optimal radii interpolate four structural regimes which we characterize exactly via Descartes’ Circle Theorem. The breakdown mechanism repeats: circle 2 leaves the chain at $n = 14$ (certified) and circle 3 by $n = 18$ (heuristic); at each of $n = 8, 13, 17$ the unrealizability of the relevant Supnick tour is itself a certified computation (global optima are certified for $n \leq 14$ only), while the indefinite continuation of the cascade is stated as an explicit conjecture. We further conjecture, with supporting analysis, that $R^*(n) = \frac{n^2}{8} (1 + o(1))$, and we determine the radius-*maximizing* arrangement as the complementary Supnick tour $\langle 1, 3, 5, \dots, n, \dots, 4, 2 \rangle$.

1 Introduction

The following question was asked by the user Dan on Mathematics Stack Exchange in January 2023 [1] and, despite substantial attention (27 votes), has remained unanswered; it appears as question 295 in the UQ benchmark of unsolved Stack Exchange questions [2].

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Non-overlapping circles of radius $1, 2, 3, \dots, n$ are all externally tangent to a middle circle. How should we arrange the surrounding circles in order to minimize the middle circle's radius R ?

Dan conjectured a specific “pyramid-fashion” arrangement and observed two phenomena that make the problem subtle: (i) the natural closure equation

$$\sum_{k=1}^n \arccos\left(1 - \frac{2r_k r_{k+1}}{(R + r_k)(R + r_{k+1})}\right) = 2\pi \quad (r_{n+1} = r_1) \quad (1)$$

treats consecutive circles as mutually tangent, yet (ii) for $n \geq 8$ the smallest circles can be “placed last anywhere they fit”, i.e. the optimal configuration contains circles tangent to the central circle only. A comment by Rei Henigman [1] verified the conjecture for $n = 10$ by enumerating permutations of $\{2, \dots, 10\}$, manually excluding circle 1 from the chain.

This paper addresses the problem on three levels, keeping the epistemic status of each result explicit.

1. The ordering problem is a Supnick TSP. Writing $\theta_R(a, b)$ for the minimum angular separation, at central radius R , between the centers of tangent circles of radii a and b (Section 2), we show that the matrix $(\theta_R(r_i, r_j))_{i,j}$ is symmetric *anti-Monge* for every $R > 0$ (Lemma 5). Minimizing total angle over cyclic orders is therefore a *maximum* traveling salesman problem on a Supnick matrix, which by a classical theorem of Supnick [3] (see also the survey [4]) is solved by one fixed tour independent of the matrix entries:

$$\sigma^* = \langle 1, n-1, 3, n-3, 5, n-5, \dots, n-2, 2, n \rangle.$$

A short self-consistency argument (the same order is optimal at *every* R) transfers optimality from fixed R to the variable- R problem (Theorem 6). The order σ^* coincides with Dan’s conjectured pyramid arrangement. In particular $R_{\text{chain}}(\sigma^*)$, defined by (1), is an unconditional lower bound for $R^*(n)$, and it is the exact optimum whenever the σ^* -necklace is geometrically realizable.

2. Realizability breaks in a cascade, which we determine exactly. The closure equation (1) ignores non-adjacent pairs. For $n \geq 8$ the configuration it describes for σ^* is *not realizable*: circle 1, seated between circles n and $n-1$, no longer keeps them apart, i.e. $\theta_R(n, 1) + \theta_R(1, n-1) < \theta_R(n, n-1)$. Circle 1 then leaves the chain and becomes a *floating* circle, tangent to the central circle only. We prove that the correct model is a circular system of $\binom{n}{2}$ pairwise angular constraints — equivalently a simple temporal network (STN) — and we give an exact, certified algorithm for it (Section 4). Exhaustive certified optimization for $3 \leq n \leq 14$ (Section 5) reveals four regimes, governed by two closed-form quantities of the Supnick tour on $\{2, \dots, n\}$: its geometric validity and the largest Descartes pocket it opens.

regime	n	floating set	optimal chain
I (full necklace)	$3 \leq n \leq 7$	$\emptyset$	σ^* on $\{1, \dots, n\}$
II (paid floater)	$n = 8, 9$	$\{1\}$	distorted tour on $\{2, \dots, n\}$
III (free floater)	$n = 10, 11, 12$	$\{1\}$	σ^* on $\{2, \dots, n\}$
IV (second breakdown)	$n = 13$	$\{1\}$	constrained tour on $\{2, \dots, n\}$
II' (cascade, level 2)	$n = 14$	$\{1, 2\}$	constrained tour on $\{3, \dots, n\}$

3. Quantitative results. We report certified optima, binding structures and floating sets for all $n \leq 14$, verified to 50 decimal digits (Table 1); the radius-maximizing (worst) arrangement, which is the *minimum-tour* counterpart $\langle 1, 3, 5, \dots, n, \dots, 4, 2 \rangle$ of Supnick’s theorem (Proposition 8); and the conjectured asymptotics $R^*(n) \sim n^2/8$ (Section 6).

Epistemic ledger. Three levels of certainty are kept explicit throughout. *Proved:* Lemmas 1 and 5, Theorem 6, and the chain-level statement of Proposition 8. *Certified computation* (exhaustive search with explicitly stated numerical guards, independently re-checkable from the repository): Tables 1 and 3, the realizability statements of Propositions 9–11 on their stated finite ranges, and the three seam-failure computations. *Heuristic or conjectural:* Table 2, Conjecture 13, and Conjecture 14.

Related work. Supnick’s fixed-tour theorem [3] and the Monge/Supnick taxonomy of polynomially solvable TSPs are surveyed in [4]. Circle packing literature concerns containers (packing circles *inside* regions, see e.g. [6]); the present problem — ordering unequal circles *around* a central one — appears not to have been treated. The tangency mechanics are governed by Descartes’ Circle Theorem [5]. To our knowledge the link between Supnick matrices and rings of tangent circles is new.

2 The model

Let the central circle have radius $R > 0$ and center O . A surrounding circle of radius r tangent externally to it has its center at distance $R + r$ from O ; its position is a single angle φ . The whole configuration is a vector of angles.

Lemma 1 (Exact angular reformulation). *Circles of radii a, b , both externally tangent to the central circle, with angular positions separated by $\Delta \in [0, \pi]$, are non-overlapping if and only if*

$$\Delta \geq \theta_R(a, b) := 2 \arcsin \sqrt{\frac{ab}{(R+a)(R+b)}}.$$

Moreover $\theta_R(a, b) \in (0, \pi)$ always, θ_R is symmetric, strictly decreasing in R , and strictly increasing in each radius.

Proof. By the law of cosines the squared distance between the centers is $(R+a)^2 + (R+b)^2 - 2(R+a)(R+b)\cos\Delta$, and non-overlap means this is at least $(a+b)^2$. Rearranging gives $\cos\Delta \leq 1 - \frac{2ab}{(R+a)(R+b)}$, i.e. $\Delta \geq \theta_R(a, b)$ with θ_R as stated, using $\arccos(1-2s^2) = 2 \arcsin s$ for $s \in [0, 1]$. The argument of the square root is strictly less than 1 since $(R+a)(R+b) - ab = R(R+a+b) > 0$, whence $\theta_R < \pi$; the monotonicity claims are immediate from the formula. $\square$

Thus the problem is: minimize R such that there exist angles $\varphi_1, \dots, \varphi_n$ with circular separations $\geq \theta_R(r_i, r_j)$ for *all* pairs. For a fixed cyclic order this is a circular system of difference constraints — a simple temporal network — and feasibility at fixed R is decidable by shortest paths (Section 4). Two derived quantities recur:

Definition 2. For a cyclic order σ of a multiset of radii, $R_{\text{chain}}(\sigma)$ is the unique solution of $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) = 2\pi$. (Existence and uniqueness: the left side tends to $|\sigma|\pi \geq 2\pi$ as $R \rightarrow 0^+$, to 0 as $R \rightarrow \infty$, and is strictly decreasing.) For a feasible pair (order, R), summing the adjacent

gaps shows $\sum_i \theta_R(\sigma_i, \sigma_{i+1}) \leq 2\pi$ and hence $R \geq R_{\text{chain}}(\sigma)$: the chain value is an unconditional per-order lower bound.

Definition 3 (Floating circle). A circle c is *floating* at a global optimum if there exists an optimal configuration in which c is tangent only to the central circle (every pairwise constraint involving c is strictly slack).

Definition 4 (Descartes pocket). For adjacent, mutually tangent circles a, b on the necklace at central radius R , the *pocket* is the largest circle externally tangent to the central circle and to both: by Descartes' Circle Theorem its curvature is $k_{\text{pocket}} = \frac{1}{R} + \frac{1}{a} + \frac{1}{b} + 2\sqrt{\frac{1}{Ra} + \frac{1}{ab} + \frac{1}{Rb}}$. A circle of radius ρ floats freely between a and b iff $\rho \leq 1/k_{\text{pocket}}$, equivalently $\theta_R(a, \rho) + \theta_R(\rho, b) \leq \theta_R(a, b)$.

3 The optimal cyclic order is a fixed Supnick tour

Lemma 5 (Supermodularity). *For every $R > 0$ the matrix $(\theta_R(r_i, r_j))$, rows and columns indexed by the radii in increasing order, is symmetric strictly anti-Monge: for $a < a'$ and $b < b'$,*

$$\theta_R(a, b) + \theta_R(a', b') > \theta_R(a, b') + \theta_R(a', b).$$

Proof. Write $\theta_R(a, b) = 2 \arcsin(t_a t_b)$ with $t_x = \sqrt{x/(R+x)} \in (0, 1)$, which is strictly increasing in x . For $f(x, y) = \arcsin(xy)$ on $(0, 1)^2$,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y} \frac{y}{\sqrt{1-x^2 y^2}} = \frac{(1-x^2 y^2) + x^2 y^2}{(1-x^2 y^2)^{3/2}} = (1-x^2 y^2)^{-3/2} > 0,$$

so f is strictly supermodular; supermodularity is preserved by strictly increasing reparametrizations of each coordinate, and the claim follows. $\square$

Equivalently, $-\theta_R$ is a Supnick matrix — symmetric Monge off the diagonal, diagonal entries being irrelevant for tours (cf. [4, Prop. 2.13]). Supnick's theorem [3] covers both extreme tours; in the form stated in [4] (Theorem 2.5 for the minimum, and the max-TSP discussion following Theorem 2.12, both attributed there to [3]), the minimum TSP tour on Supnick matrices is $\langle 1, 3, 5, \dots, n, \dots, 6, 4, 2 \rangle$ and the *maximum* TSP tour is

$$\sigma^* = \langle 1, n-1, 3, n-3, 5, n-5, \dots, n-2, 2, n \rangle,$$

independently of the matrix entries. Maximizing $\sum(-\theta_R)$ is minimizing $\sum \theta_R$, hence:

Theorem 6 (Optimal order). *Let $r_1 < \dots < r_n$ be arbitrary positive radii.*

- (i) *For every $R > 0$, among all cyclic orders the tour σ^* minimizes $\sum_i \theta_R(r_{\sigma(i)}, r_{\sigma(i+1)})$.*
- (ii) *σ^* minimizes R_{chain} over all cyclic orders; hence $R^* \geq R_{\text{chain}}(\sigma^*)$ unconditionally.*
- (iii) *If the σ^* -necklace at $R_{\text{chain}}(\sigma^*)$ is geometrically realizable (no non-adjacent pair violated), then it is a global optimum: $R^* = R_{\text{chain}}(\sigma^*)$.*

Proof. (i) is Lemma 5 plus Supnick's theorem. (ii): fix any order σ and let $R = R_{\text{chain}}(\sigma)$. By (i), $\sum \theta_R(\sigma^*) \leq \sum \theta_R(\sigma) = 2\pi$; since $\sum \theta_R(\sigma^*)$ is strictly decreasing in R and equals 2π at $R_{\text{chain}}(\sigma^*)$, we get $R_{\text{chain}}(\sigma^*) \leq R$. The unconditional bound follows since every feasible (σ, R) has $R \geq R_{\text{chain}}(\sigma)$. (iii): realizability makes the lower bound attained. $\square$

Remark 7. Theorem 6 holds for *arbitrary* distinct radii, not only $1, \dots, n$; the arithmetic sequence enters only in the realizability analysis below. For $n = 10$ the lower bound of (ii) is ≈ 9.90 , the value observed in [1] when circle 1 is (incorrectly) kept in the chain; the true optimum is $9.9799\dots$

Proposition 8 (Worst arrangement). *For every R the cyclic order maximizing $\sum_i \theta_R$ is the complementary Supnick tour $\sigma_{\dagger} = \langle 1, 3, 5, \dots, n, \dots, 6, 4, 2 \rangle$; consequently $\sigma_{\dagger}$ maximizes R_{chain} . For all $3 \leq n \leq 13$ the $\sigma_{\dagger}$ -necklace is realizable, so among necklace arrangements it maximizes the central radius exactly; certified values in Table 3.*

4 Exact algorithm and certification

Feasibility oracle. Fix a cyclic order and R . Unrolling the circle at one circle ($\varphi_0 = 0$) gives, for every pair $i < j$ of positions, the two constraints $\varphi_j - \varphi_i \geq \theta_R(r_i, r_j)$ and $\varphi_j - \varphi_i \leq 2\pi - \theta_R(r_i, r_j)$. This is a simple temporal network; feasibility is the absence of a negative cycle in the associated distance graph and is decided by Floyd–Warshall in $O(n^3)$. The minimum feasible R for the order, $R_{\text{full}}(\sigma)$, is found by bisection (the constraint set relaxes monotonically in R), bracketed below by $R_{\text{chain}}(\sigma)$. A witness configuration is recovered from shortest-path potentials and verified independently in Cartesian coordinates.

Lower bounds and search. For the search over orders we use

$$\text{LB}(\sigma) = \max \{ R_{\text{chain}}(\sigma), R_{\text{chain}}(\sigma \setminus \{1\}), R_{\text{chain}}(\sigma \setminus \{1, 2\}) \},$$

where $\sigma \setminus S$ denotes the induced cyclic order (valid since induced adjacencies are a subset of the full pairwise constraints). Orders are enumerated canonically (largest radius pinned, reflections halved), LB is computed vectorized for all $(n-1)!/2$ orders, and R_{full} is evaluated in ascending LB until the next bound meets the incumbent: this certifies global optimality up to an absolute guard of 10^{-10} in R . The guard dominates the floating-point error budget: bisections run to relative tolerance 10^{-13} ; one bound evaluation is an n -term sum of arcsin values, with relative error $O(n \varepsilon_{64})$; and the float64 bounds were calibrated against 50-digit recomputation on random samples of orders, with maximal observed deviation 1.8×10^{-14} (over 10^5 random orders for each $8 \leq n \leq 14$; maximal overestimate 1.5×10^{-14}), four orders of magnitude below the guard. Independently of the search code, the repository ships a standalone verifier (`verify.py`) that re-derives each claimed optimum and re-checks every order on the pruning frontier (between 1 and 11 orders per n) in 50-digit arithmetic. Optimal binding structures were re-verified in 50-digit arithmetic (mpmath); the values in Table 1 are exact to all displayed digits, with maximal residual slack below 10^{-40} . The $n = 13$ certification enumerates $12!/2 \approx 2.4 \times 10^8$ orders (≈ 1.8 h on a consumer 8-core laptop); $n = 14$ enumerates $13!/2 \approx 3.1 \times 10^9$ orders (≈ 22 h). Code and artifacts: <https://github.com/falker47/ringmin>.

Essential binding structure. A pairwise constraint is *essential* if perturbing it alone by $+\varepsilon$ makes the STN infeasible at R^* ; a circle is floating iff it carries no essential constraint. This distinguishes true tangencies from placement artifacts of a particular witness.

Table 1: Certified optima, rounded to 12 decimals from the 50-digit verified values (30-digit table in the Appendix). Cycles list radii in circular order; the floating circle is shown in its (non-unique) hosting gap. “Essential pair” is the tangent non-adjacent pair bridged by the floater.

n	$R^*(n)$	optimal cycle	floating	essential pair
3	0.260869565217 (= 6/23)	(3,1,2)	—	—
4	0.844453589561	(4,1,3,2)	—	—
5	1.695494081203	(5,1,4,3,2)	—	—
6	2.794919518897	(6,1,5,3,4,2)	—	—
7	4.153189553744	(7,1,6,3,4,5,2)	—	—
8	5.767794284590	(8,1,6,4,5,3,7,2)	{1}	(8, 6)
9	7.726726552611	(9,2,8,1,5,6,4,7,3)	{1}	(8, 5)
10	9.979907385863	(10,2,9,4,7,1,6,5,8,3)	{1}	(7, 6)
11	12.488720487188	(11,2,10,4,8,6,7,1,5,9,3)	{1}	(7, 5)
12	15.258870430448	(12,2,11,4,9,6,7,8,5,1,10,3)	{1}	(5, 10)
13	18.317563047217	(13,3,1,12,2,10,6,8,7,9,5,11,4)	{1}	(3, 12)
14	21.665395182215	(14,3,13,2,9,8,7,10,6,11,5,1,12,4)	{1, 2}	(13, 9), (5, 12)

5 Certified optima for $n \leq 14$ and the regime cascade

For $n \leq 7$ the optimal cycle is exactly σ^* on $\{1, \dots, n\}$ (regime I), as guaranteed by Theorem 6(iii) once realizability is checked. The transitions are governed by two computable properties of the Supnick tour on the reduced set $\{2, \dots, n\}$:

Proposition 9 (First breakdown). *In the σ^* -necklace on $\{1, \dots, n\}$, circle 1 sits between n and $n - 1$. The realizability inequality $\theta_R(n, 1) + \theta_R(1, n - 1) \geq \theta_R(n, n - 1)$ at $R = R_{\text{chain}}(\sigma^*)$ holds for $n \leq 7$ and fails for $8 \leq n \leq 13$ (violated pair $(n, n - 1)$; at $n = 13$ also $(13, 11)$ across circle 2). Hence regime I ends at $n = 7$. We conjecture failure for all $n \geq 8$; cf. open problem (1).*

Proposition 10 (Free vs. paid floater). *Let $\rho_{\max}(n)$ be the largest Descartes pocket of the σ^* -necklace on $\{2, \dots, n\}$ at its chain radius. Then $\rho_{\max} = 0.858, 0.988, 1.139, 1.262, 1.409$ for $n = 8, \dots, 12$. For $n = 10, 11, 12$ one has $\rho_{\max} > 1$: circle 1 floats freely and $R^*(n) = R_{\text{chain}}(\sigma^* \text{ on } \{2, \dots, n\})$ exactly (regime III). For $n = 8, 9$, $\rho_{\max} < 1$: the optimum distorts the reduced tour to open a unit pocket, at certified extra cost $\delta(8) = 0.04080$, $\delta(9) = 0.00532$ over the (then unattainable) reduced chain bound (regime II).*

Proposition 11 (Second breakdown). *At $n = 13$ the Supnick tour on $\{2, \dots, 13\}$ is itself unrealizable: circle 2, seated between 13 and 12, violates $\theta_R(13, 2) + \theta_R(2, 12) = 0.80448 < 0.83488 = \theta_R(13, 12)$ at $R = 18.3176$. The certified optimum demotes circle 2 to the pair $(12, 10)$, where the separation constraint holds with slack 6×10^{-3} rad; circle 1 floats in the $(3, 12)$ pocket (radius 1.086). Thus the mechanism that expelled circle 1 at $n = 8$ reaches circle 2 at $n = 13$ (regime IV).*

Remark 12 (Structural summary). For all $n \leq 14$ the optimum has the form: a single tangent necklace carrying a Supnick-like order *constrained to geometric realizability*, plus floating circles in Descartes pockets. The cascade continues self-similarly: circle 2 first floats at $n = 14$ (now *certified*, Table 1) via a *paid* distortion that opens a pocket of radius 2.013 between circles 13 and 9 — the regime-II mechanism one level up, at extra cost 0.0565 over the (then unattainable) chain bound of the Supnick tour on $\{3, \dots, 14\}$. At $n = 16$ the best known value coincides to

10^{-10} with the chain value of the Supnick tour on $\{3, \dots, 16\}$, the level-2 analogue of regime III. Heuristically, circle 3 joins by $n = 18$ (Table 2). Moreover the Supnick tour on $\{3, \dots, n\}$ itself becomes unrealizable at $n = 17$ — circle 3 failing at the $(17, 16)$ seam with slack -0.0104 — so the *seam-failure* onsets for circles 1, 2, 3 are $n = 8, 13, 17$: each *computed* cascade level reproduces the full regime pattern of the previous one. The float onsets $n = 8, 14, 18$ for circles 1, 2, 3 are consistent with Dan’s $n/4$ Descartes-based upper bound on the number of ignorable circles [1]. Whether the cascade continues indefinitely is the content of the following conjecture.

Conjecture 13 (Cascade). *For every $k \geq 1$ there is a threshold s_k such that the Supnick tour on $\{k, \dots, n\}$ is geometrically unrealizable for all $n \geq s_k$ (observed and certified: $s_1 = 8$, $s_2 = 13$, $s_3 = 17$), and circle k is floating in every optimal configuration for all sufficiently large n , with each level exhibiting the paid-then-free regime pattern of Propositions 9–11.*

Table 2: Best known arrangements for $15 \leq n \leq 18$ (local search, 200–600 restarts per n ; *non-exhaustive*). The last column counts restarts reaching the best value.

n	best R	floating	cycle	hits
15	25.290151636714	$\{1, 2\}$	(15,3,14,5,12,2,9,1,8,10,7,11,6,13,4)	59
16	29.194988498141	$\{1, 2\}$	(16,1,4,14,6,12,8,10,9,2,11,7,13,5,15,3)	18
17	33.394425022062	$\{1, 2\}$	(17,4,16,3,15,6,1,13,8,11,10,9,2,12,7,14,5)	19
18	37.870755667393	$\{1, 2, 3\}$	(18,4,17,3,14,8,12,10,11,9,2,13,7,15,6,16,1,5)	9

6 Asymptotics

For $R \gg n$ one has $\theta_R(a, b) = 2\sqrt{ab}/R(1 + O(n/R))$, so the closure equation gives $R \approx \frac{1}{\pi} \sum_i \sqrt{r_i r_{i+1}}$. Since $\sqrt{ab}$ is supermodular, Theorem 6(i) applies verbatim: the minimizing order of $\sum \sqrt{r_i r_{i+1}}$ is again σ^* , whose value on $1, \dots, n$ is

$$\min_{\sigma} \sum_i \sqrt{r_{\sigma(i)} r_{\sigma(i+1)}} = \left(1 + o(1)\right) \int_0^n \sqrt{x(n-x)} dx = \frac{\pi n^2}{8} (1 + o(1)),$$

because adjacent pairs in σ^* are $\{k, n - k + O(1)\}$ with each odd k contributing twice. Self-consistency ($R \sim n^2/8 \gg n$) further requires the floating circles to be asymptotically negligible, in the sense $\sum_{k \in F(n)} \sqrt{k n} = o(n^2)$ — guaranteed if $|F(n)| = o(n)$, consistent with $|F(18)| = 3$, but open (problem (2) below). Under this assumption we are led to:

Conjecture 14. $R^*(n) = \frac{n^2}{8}(1 + o(1))$. *Empirically the deficit $n^2/8 - R(n)$ equals 2.52, 2.64, 2.74, 2.81, 2.83 for the certified $n = 10, \dots, 14$, and 2.83, 2.81, 2.73, 2.63 for the heuristic $n = 15, \dots, 18$ (the latter are lower bounds on the true deficit, heuristic radii being upper bounds for R^*): bounded so far, and in any case growing strictly slower than n . We tentatively conjecture the stronger form $n^2/8 - R^*(n) = O(\sqrt{n})$.*

Two-sided bounds making the leading order rigorous (via $\arcsin x \geq x$ and the explicit σ^* configuration as an upper bound) appear within reach but are not pursued here.

7 Open problems

(1) Prove Propositions 9–11 for all n analytically (the inequalities are explicit; monotonicity in n is the missing step). (2) Characterize the floating set $F(n)$ asymptotically (float onsets: circle 1 at $n = 8$ and circle 2 at $n = 14$, both certified, circle 3 at $n = 18$ heuristically; certified seam-failure onsets $n = 8, 13, 17$); is $|F(n)| \sim cn^{1/2}$, or does Dan’s $n/4$ bound give the right order? Both are subsumed by proving Conjecture 13. (3) The same machinery applies to radii $r_k = k^\alpha$ or arbitrary sequences: Theorem 6 is general, the cascade analysis is not. (4) Three dimensions: spheres of radii $1, \dots, n$ tangent to a central sphere — the ordering structure disappears; what replaces Supnick?

Acknowledgments

The problem and the pyramid conjecture are due to Dan [1]; Rei Henigman’s $n = 10$ computation provided an independent check. This work was produced with substantial AI assistance — research design, mathematical arguments, and software — under the author’s direction and final review; a detailed account of the workflow is given in the repository README. The paper is written to be independently checkable: the proof of Theorem 6 is short and self-contained, the only imported result is Supnick’s theorem, and every certified value is reproducible from the public repository, independently of any AI system (Section 4).

References

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- [4] R.E. Burkard, V.G. Deineko, R. van Dal, J.A.A. van der Veen, G.J. Woeginger, *Well-solvable special cases of the TSP: a survey*, SIAM Review **40**(3) (1998), 496–546.
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A High-precision values and figures

Per- n configuration figures for all $3 \leq n \leq 18$ are available in the repository.

Table 3: The radius-maximizing necklace arrangement $\sigma_{\dagger}$ (Proposition 8): certified chain values, all geometrically realizable.

n	3	4	5	6	7	8	9	10	11	12	13
$R_{\dagger}$	0.2609	0.8673	1.7786	3.0001	4.5375	6.3934	8.5690	11.0647	13.8806	17.0164	20.4720

n	R^*
3	0.260869565217391304347826086957
4	0.844453589560855604347528524674
5	1.69549408120271081351328017371
6	2.79491951889692485617024406797
7	4.15318955374381246513863858202
8	5.7677942845896143026361805725
9	7.72672655261128921886246177604
10	9.97990738586347760966552641468
11	12.4887204871876588517468786264
12	15.2588704304484933617250043503
13	18.3175630472173206282821941532
14	21.6653951822145150956462891793

Table 4: Certified optima, 30 significant digits.

n	Supnick min tour	R_{chain}	R_{full}
3	[1, 3, 2]	0.260869565217	0.260869565217
4	[1, 3, 4, 2]	0.867281640255	0.867281640255
5	[1, 3, 5, 4, 2]	1.778550949884	1.778550949884
6	[1, 3, 5, 6, 4, 2]	3.000138442323	3.000138442323
7	[1, 3, 5, 7, 6, 4, 2]	4.537467833562	4.537467833562
8	[1, 3, 5, 7, 8, 6, 4, 2]	6.393358320114	6.393358320114
9	[1, 3, 5, 7, 9, 8, 6, 4, 2]	8.568988935955	8.568988935955
10	[1, 3, 5, 7, 9, 10, 8, 6, 4, 2]	11.064726494858	11.064726494858
11	[1, 3, 5, 7, 9, 11, 10, 8, 6, 4, 2]	13.880577932626	13.880577932626
12	[1, 3, 5, 7, 9, 11, 12, 10, 8, 6, 4, 2]	17.016408238128	17.016408238128
13	[1, 3, 5, 7, 9, 11, 13, 12, 10, 8, 6, 4, 2]	20.472039120681	20.472039120681
14	[1, 3, 5, 7, 9, 11, 13, 14, 12, 10, 8, 6, 4, 2]	24.247291190092	24.247291190092
15	[1, 3, 5, 7, 9, 11, 13, 15, 14, 12, 10, 8, 6, 4, 2]	28.342000288518	28.342000288518
16	[1, 3, 5, 7, 9, 11, 13, 15, 16, 14, 12, 10, 8, 6, 4, 2]	32.756022334220	32.756022334220
17	[1, 3, 5, 7, 9, 11, 13, 15, 17, 16, 14, 12, 10, 8, 6, 4, 2]	37.489233291550	37.489233291550
18	[1, 3, 5, 7, 9, 11, 13, 15, 17, 18, 16, 14, 12, 10, 8, 6, 4, 2]	42.541527273519	42.541527273519

Table 5: Worst-arrangement comparison; Supnick min tour values.

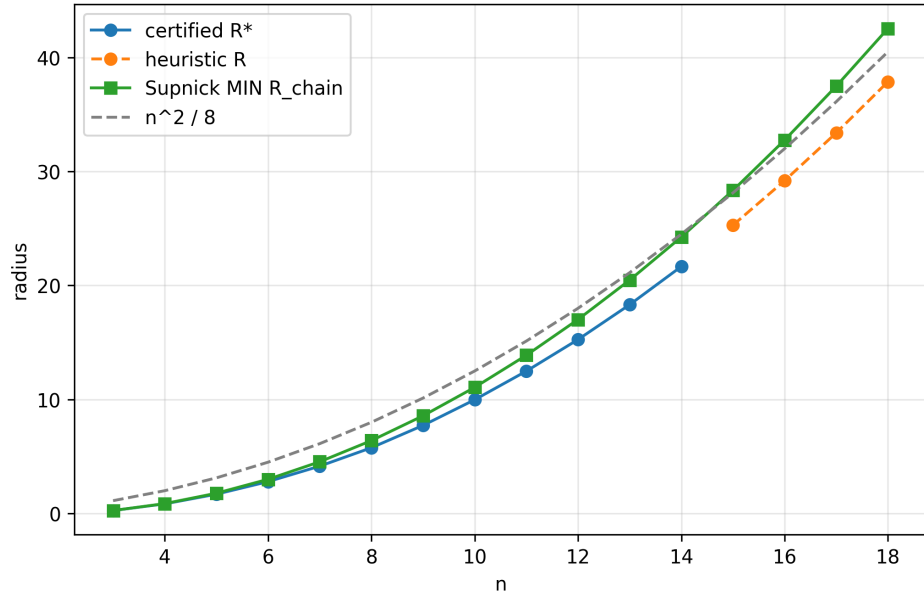


Figure 1: $R^*(n)$, the worst-arrangement curve, and the $n^2/8$ asymptote.

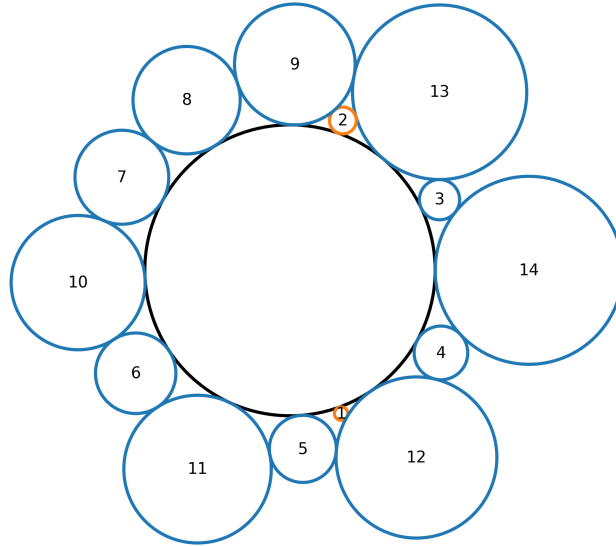


Figure 2: The certified optimum for $n = 14$; floating circles 1, 2 highlighted.